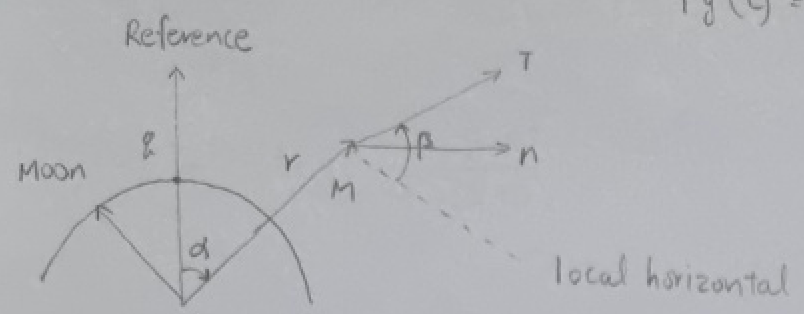
최적제어 HW #5 2022311224 이진진

ex 5.1-2)

$F_g(t) = Mg \cdot R^2 / r^2(t)$



$x_1 \triangleq r$, $x_2 \triangleq \alpha$, $x_3 \triangleq \dot{r}$ $x_4 \triangleq r\dot{\alpha}$, $u \triangleq \beta$ 라고 하면

$$\dot{x}_1(t) = x_3(t)$$

$$\dot{x}_2(t) = \frac{x_4(t)}{x_1(t)}$$

$$\dot{x}_3(t) = \frac{x_4^2(t)}{x_1(t)} - \frac{gR^2}{x_1(t)} + \left[\frac{T}{M}\right] \sin u(t)$$

$$\dot{x}_4(t) = -\frac{x_3(t) x_4(t)}{x_1(t)} + \left[\frac{T}{M}\right] \cos u(t)$$

⟨Mission a⟩

$$H = 1 + P_1(t) x_3(t) + \frac{P_2(t) x_4(t)}{x_1(t)} + P_3(t)\left[\frac{x_4^2(t)}{x_1(t)} - \frac{gR^2}{x_1^2(t)} + \left[\frac{T}{M}\right]\sin u(t)\right] + P_4(t)\left[-\frac{x_3(t)x_4(t)}{x_1(t)} + \left[\frac{T}{M}\right]\cos u(t)\right]$$

$$\dot{P}_1^*(t) = -\frac{\partial H}{\partial x_1} = \frac{P_2^*(t) x_4^*(t)}{x_1^{*2}(t)} + P_3^*(t)\left[\frac{x_4^{*2}(t)}{x_1^{*2}(t)} - \frac{2gR^2}{x_1^{*3}(t)}\right] - \frac{P_4^*(t) x_3^*(t) x_4^*(t)}{x_1^{*2}(t)}$$

$$\dot{P}_2^*(t) = -\frac{\partial H}{\partial x_2} = 0$$

$$\dot{P}_3^*(t) = -\frac{\partial H}{\partial x_3} = -P_1^*(t) + \frac{P_4^*(t) x_4^*(t)}{x_1^*(t)}$$

$$\dot{P}_4^*(t) = -\frac{\partial H}{\partial x_4} = -\frac{P_2^*(t)}{x_1^*(t)} - \frac{2P_3^*(t) x_4^*(t)}{x_1^*(t)} + \frac{P_4^*(t) x_3^*(t)}{x_1^*(t)}$$

State Eq. : $\dot{x}^*(t) = \vec{u}(\vec{x}^*(t), u^*(t))$

$$0 = \frac{\partial H}{\partial u} = \left[\frac{T}{M}\right]\left[P_3^*(t)\cos u^*(t) - P_4^*(t)\sin u^*(t)\right] \Rightarrow u^*(t) = \tan^{-1}\frac{P_3^*(t)}{P_4^*(t)} \quad \therefore \left[\begin{array}{l} \sin u^*(t) = \dfrac{P_3^*(t)}{\sqrt{P_3^{*2}(t) + P_4^{*2}(t)}} \\ \cos u^*(t) = \dfrac{P_4^*(t)}{\sqrt{P_3^{*2}(t) + P_4^{*2}(t)}} \end{array}\right.$$

B.C : $x_1^*(t_f) = R+D$.

$p_2^*(t_f) = 0$ $\qquad H(\vec{x}^*(t_f), \vec{p}^*(t_f)) = 0$

$x_3^*(t_f) = 0$

$x_4^*(t_f) = \sqrt{\dfrac{g \cdot R^2}{(R+D)}}$

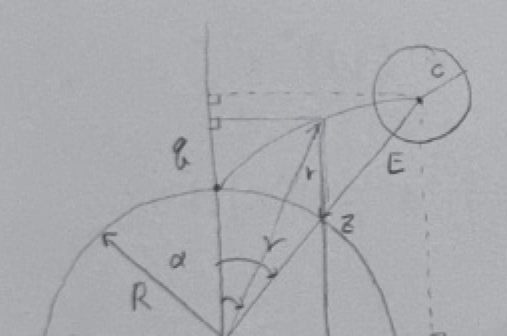
⟨Mission b⟩

$$\theta(t) = \begin{bmatrix} R+D \\ \pi t_f - 2\pi n \\ 0 \\ \pi(R+D) \end{bmatrix} \qquad \delta x_{2f} = \left[\frac{d\theta_2}{dt}(t_f)\right]\delta t_f = \pi \delta t_f$$

$$-\pi p_2^*(t_f) + H(\vec{x}^*(t_f), \vec{p}^*(t_f)) = 0$$

B.C : $\vec{x}^*(t_f) = \begin{bmatrix} R+D \\ \pi t_f - 2\pi n \\ 0 \\ \pi(R+D) \end{bmatrix} = \theta(t_f)$

⟨Mission C⟩



$$m(\vec{x}(t)) = [r(t)\cos\alpha(t) - (R+E)\cos\gamma]^2 + [r(t)\sin\alpha(t) - (R+E)\sin\gamma]^2 - C^2 = 0$$

$$H + \frac{\partial h}{\partial t} = \sum_{i=1}^{k} d_i \left[\frac{\partial m_i}{\partial t}(\vec{x}^*(t_f), t_f)\right]$$

$$\Rightarrow -\vec{p}^*(t_f) = d\left[\frac{\partial m}{\partial \vec{x}}(\vec{x}^*(t_f))\right]$$

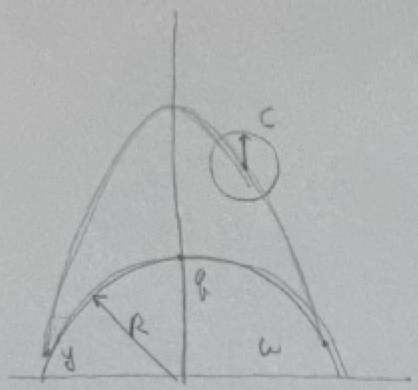
$$= d \begin{bmatrix} 2x_1^*(t_f) - 2(R+E)\cos(x_2^*(t_f) - \gamma) \\ 2x_1^*(t_f)(R+E)\sin(x_2^*(t_f) - \gamma) \\ 0 \\ 0 \end{bmatrix}$$

B.C : $p_3^*(t_f) = 0$, $p_4^*(t_f) = 0$

$$m(\vec{x}^*(t_f)) = [x_1^*(t_f)\cos x_2^*(t_f) - (R+E)\cos\gamma]^2 + [x_1^*(t_f)\sin x_2^*(t_f) - (R+E)\sin\gamma]^2 - C^2 = 0$$

$H(\vec{x}^*(t_f), p^*(t_f)) = 0$

<mission d>



$$m(\vec{x}(t), t) = [r(t)\cos\alpha(t) - 2.78Rt + 6.95Rt^2 - R]^2$$
$$+ [r(t)\sin\alpha(t) - 1.85Rt + 0.32R]^2 - c^2 = 0$$

$$-\vec{P}^*(t_f) = d\left[\frac{\partial m}{\partial \vec{x}}(\vec{x}^*(t_f), t_f)\right]$$

B.C : $-P_1^*(t_f) = 2d\left[x_1^*(t_f) + R\left\{[-2.78t_f + 6.95t_f^2 - 1]\cos x_2^*(t_f) + [-1.85t_f + 0.32]\sin x_2^*(t_f)\right\}\right]$

$-P_2^*(t_f) = -2d\left[Rx_1^*(t_f)\left\{[-2.78t_f + 6.95t_f^2 - 1]\sin x_2^*(t_f) + [1.85t_f - 0.32]\cos x_2^*(t_f)\right\}\right]$

$-P_3^*(t_f) = 0$

$-P_4^*(t_f) =$

$H(x^*(t_f), P^*(t_f)) = 2dR\{[-2.78 + 13.9t_f][x_1^*(t_f)\cos x_2^*(t_f) - 2.78Rt_f + 6.95Rt_f^2 - R]$
$- 1.85[x_1^*(t_f)\sin x_2^*(t_f) - 1.85Rt_f + 0.32R]\}$

ex 5.2-1)

$\dot{x}(t) = ax(t) + u(t)$ $\quad J(u) = \frac{1}{2}Hx^2(T) + \int_0^T \frac{1}{4}u^2(t)\,dt$ $\quad H = \frac{1}{4}u^2(t) + p^*(t)(ax(t) + u(t))$

$\dot{p}^*(t) = -\frac{\partial H}{\partial x} = -ap^*(t)$

$$\begin{bmatrix} \dot{x}^*(t) \\ \dot{p}^*(t) \end{bmatrix} = \begin{bmatrix} a & -2 \\ 0 & -a \end{bmatrix}\begin{bmatrix} x^*(t) \\ p^*(t) \end{bmatrix}$$

$$\phi(s) = \begin{bmatrix} s-a & 2 \\ 0 & s+a \end{bmatrix}^{-1} = \frac{1}{(s-a)(s+a)}\begin{pmatrix} s+a & -2 \\ 0 & s-a \end{pmatrix} = \begin{bmatrix} \frac{1}{s-a} & \frac{1}{a}\left(\frac{1}{s+a} - \frac{1}{s-a}\right) \\ 0 & \frac{1}{s+a} \end{bmatrix}$$

$$\phi(t) = \begin{bmatrix} e^{at} & \frac{1}{a}e^{-at} - \frac{1}{a}e^{at} \\ 0 & e^{-at} \end{bmatrix}$$

$k(t) = \left[e^{-a(T-t)} - \frac{H}{a}\left(e^{-a(T-t)} - e^{a(T-t)}\right)\right]^{-1}\left[He^{a(T-t)}\right]$

$\therefore u^*(t) = -2k(t)x(t)$

3)

$$J_a = h_1(\vec{x}(t_0), t_0) + h_2(\vec{x}(t_f), t_f) + \int_{t_0}^{t_f} [g + \vec{\lambda}^T(\vec{f} - \dot{\vec{x}})]\, dt$$ ($\vec{\lambda}(t)$: Lagrange Multipliers)

$$= \frac{\partial h_1}{\partial \vec{x}(t_0)} \delta \vec{x}(t_0) + \frac{\partial h_2}{\partial \vec{x}} \delta \vec{x}(t_f) - \vec{\lambda}^T \delta \vec{x} \Big|_{t_0}^{t_f}$$

$$+ \int_{t_0}^{t_f} \left[\left(\frac{\partial g}{\partial \vec{x}} + \vec{\lambda}^T \frac{\partial \vec{f}}{\partial \vec{x}} + \dot{\vec{\lambda}}^T \right) \delta \vec{x} + \left(\frac{\partial g}{\partial \vec{u}} + \vec{\lambda}^T \frac{\partial \vec{f}}{\partial \vec{u}} \right) \delta \vec{u} + (\vec{f} - \dot{\vec{x}})^T \delta \vec{\lambda} \right] dt = 0$$

$H = g + \vec{\lambda}^T \vec{f}$ 라고 하면 $\frac{\partial H}{\partial \vec{u}} = 0$, $\dot{\vec{\lambda}} = \frac{-\partial H^T}{\partial \vec{x}}$, $\dot{\vec{x}} = \frac{\partial H^T}{\partial \vec{\lambda}} = \vec{f}$

when $i = 1, 2, \cdots, k$; $\vec{x}^*(t_0) = \vec{x}_0$

($\vec{x} = x_1, x_2, \cdots x_k$)

$$\frac{\partial h_2}{\partial x}(\vec{x}((t_f), t_f) - \vec{\lambda}^*(t_f) = \sum_{i=1}^{k} \nu_i \left[\frac{\partial x_i}{\partial \vec{x}} (\vec{x}^*((t_f), t_f)) \right]$$

$$x(\vec{x}(t_f), t_f) = 0$$

$$H(\vec{x}^*(t_f), \vec{u}^*(t_f), \vec{\lambda}^*(t_f)) + \frac{\partial h_2}{\partial t}(\vec{x}^*(t_f), t_f)$$

$$= \sum_{i=1}^{k} \nu_i \left[\frac{\partial x_i}{\partial \vec{x}} (\vec{x}^*(t_f), t_f) \right]$$

when $j = k+1, k+2, \cdots, n$; $\vec{x}^*(t_f) = \vec{x}_f$

($\vec{x} = x_{k+1}, x_{k+2} \cdots, x_n$)

$$H(\vec{x}^*(t_f), \vec{u}(t_f), \vec{\lambda}^*(t_f), t_f) + \frac{\partial h_2}{\partial t}(\vec{x}^*(t_f), t_f) = 0$$

$(\delta \vec{x}(t_0) \neq 0)$

$$\frac{\partial h_1}{\partial \vec{x}}(\vec{x}^*(t_0), t_0) + \vec{\lambda}^*(t_0) = 0$$